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Publications

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Underline indicates mentee; ** undergraduate student; * corresponding

A. Journal Articles

1. **Hossain, M. J.**, Towhid, S.T., Sultana, S., Mukta, S.A., Gulshan, R., Miah, M.S. (2022) "Knowledge and Attitudes towards Thalassemia among Public University Students" *Microbial Bioactives*. 5(2). [10.25163/microbbioacts.526325](#) Survey-based study on thalassemia awareness among the Bangladeshi youth.
2. Towhid, S.T., **Hossain, M. J.**, Sammo, M.A.S., & Akter, S. (2022) "Perception of Students on Antibiotic Resistance and Prevention" *Eur. J. Bio. Biotech*. 3(3). [10.24018/ejbio.2022.3.3.341](#) Investigates student awareness and attitudes toward the antibiotic resistance crisis.
3. ***Hossain, M. J.**, Islam, M. W., Munni, U. R., Gulshan, R., Mukta, S. A., Miah, M. S., Sultana, S., Karmakar, M., Ferdous, J., & Islam, M. A. (2023) "Health-related quality of life among thalassemia patients in Bangladesh (SF-36)" *Scientific Reports*. 13(1). [10.1038/s41598-023-34205-9](#) Assesses the physical and mental health burden on patients living with thalassemia.
4. Akter, M. M., & ***Hossain, M. J.** (2024) "Food consumption patterns and sedentary behaviors among university students" *Health Science Reports*. 7, e70259. [10.1002/hsr2.70259](#) Links dietary habits and lack of activity to health risks in students.
5. ***Hossain, M. J.**, Azad, A. K., Shahid, M. S. B., Shahjahan, M., & Ferdous, J. (2024) "Prevalence, antibiotic resistance pattern for bacteriuria from patients with UTIs" *Health Science Reports*. 7, e2039. [10.1002/hsr2.2039](#) Maps antibiotic resistance landscapes for UTI-causing bacteria in clinical settings.
6. ***Hossain, M. J.**, Das, M., Islam, M. W., Shahjahan, M., & Ferdous, J. (2024) "Community engagement and social participation in dengue prevention" *Health Science Reports*. 7, e2022. [10.1002/hsr2.2022](#) Evaluates how social participation impacts the effectiveness of dengue control.
7. Islam, M. W., Shahjahan, M., Azad, A. K., & ***Hossain, M. J.** (2024) "Factors contributing to antibiotic misuse among parents in Dhaka City" *Scientific Reports*. 14, 2318. [10.1038/s41598-024-52313-y](#) Identifies parental socio-economic factors driving antibiotic misuse in children.
8. ***Hossain, M. J.**, Das, M., & Munni, U. R. (2024) "Urgent call for compulsory premarital screening for thalassemia prevention in Bangladesh" *Orphanet Journal of Rare Diseases*. 19, 326. [10.1186/s13023-024-03344-1](#) Advocates for policy-driven premarital screening to reduce thalassemia incidence.

9. ***Hossain, M. J.**, Sony, S. A., Fariha, F. T. J., & Hossen, S. (2025) "Preventing the silent threat: A perspective on preparing Bangladesh for Human Metapneumovirus" *Health Science Reports*. 8, e71101. [10.1002/hsr2.71101](https://doi.org/10.1002/hsr2.71101) Strategic perspective on preparing health systems for HMPV outbreaks in Bangladesh.
10. Das, M., & **Hossain, M. J.** (2025) "Young stroke in Bangladesh: Addressing rare cases with diagnostic challenges" *Stroke and Vascular Neurology*. svn-2025-004178. [10.1136/svn-2025-004178](https://doi.org/10.1136/svn-2025-004178) Call for action on diagnosing and managing rare stroke cases in young Bangladeshi patients.
11. ***Hossain, M. J.**, Das, M., Shahjahan, M., Islam, M. W., & Towhid, S. T. (2025) "Clinical and hematological manifestation of dengue patients in 2022 outbreak" *Health Science Reports*. 8, e70356. [10.1002/hsr2.70356](https://doi.org/10.1002/hsr2.70356) Study details clinical profiles and blood changes during a major Dhaka dengue outbreak.
12. Shanta, A. S., Islam, N., Al Asad, M., Akter, K., Habib, M. B., **Hossain, M. J.**, Nahar, S., Godman, B., & Islam, S. (2024) "Resistance and co-resistance of metallo-beta-lactamase genes in diarrheal and urinary-tract pathogens" *Microorganisms*. 12(8), 1589. [10.3390/microorganisms12081589](https://doi.org/10.3390/microorganisms12081589) High prevalence of MBL genes found in clinical pathogens within Bangladesh.
13. Bari, S. M., Fuad, M., **Hossain, M. J.**, Ahammad, I., Begum, M., & Helal, M. M. U. (2025) "A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout" *BMC Genomics*. 26, 999. [10.1186/s12864-025-12127-2](https://doi.org/10.1186/s12864-025-12127-2) Meta-analysis reveals conserved transcriptomic signatures of stress in rainbow trout.
14. Ahmed, M. Z., Billah, M. M., Ferdous, J., & ***Hossain, M. J.** (2025) "Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors" *Scientific Reports*. 15, 14405. [10.1038/s41598-025-88339-z](https://doi.org/10.1038/s41598-025-88339-z) CCT5 identified as a pan-cancer biomarker linked to immune infiltration and prognosis.
15. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & ***Hossain, M. J.** (2025) "Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer" *Scientific Reports*. 15, 34490. [10.1038/s41598-025-13315-6](https://doi.org/10.1038/s41598-025-13315-6) WNT genes serve as critical prognostic biomarkers and immune regulators in breast cancer.

B. GBD Contributions

1. GBD 2023 Mental Disorder Collaborators (2026) "Updated trends in the global prevalence and burden of mental disorders, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023" *The Lancet*. [https://doi.org/10.1016/S0140-6736\(26\)00519-2](https://doi.org/10.1016/S0140-6736(26)00519-2)
2. GBD 2023 Breast Cancer Collaborators (2026) "Global, regional, and national burden of breast cancer among females, 1990–2023, with forecasts to 2050: a systematic analysis for the Global Burden of Disease Study 2023" *The Lancet Oncology*. [https://doi.org/10.1016/S1470-2045\(25\)00730-2](https://doi.org/10.1016/S1470-2045(25)00730-2)
3. GBD 2023 Headache Collaborators (2025) "Global, regional, and national burden of headache disorders, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023" *The Lancet Neurology*. [https://doi.org/10.1016/S1474-4422\(25\)00402-8](https://doi.org/10.1016/S1474-4422(25)00402-8)
4. GBD 2023 Vaccine Coverage Collaborators (2025) "Global, regional, and national trends in routine childhood vaccination coverage from 1980 to 2023 with forecasts to 2030: A systematic analysis for the Global Burden of Disease Study 2023" *The Lancet*. [https://doi.org/10.1016/S0140-6736\(25\)01037-2](https://doi.org/10.1016/S0140-6736(25)01037-2)
5. GBD 2021 Musculoskeletal Disorders and COVID-19 Collaborators (2025) "Global, regional, and national burden and trend of adult musculoskeletal disorders and discrepancy between pre-pandemic and pandemic of COVID-19" *BMC Medicine (Under Review)*.
6. GBD 2023 TB Collaborators (2025) "Global, regional, and national burden of tuberculosis and multidrug-resistant tuberculosis by HIV status, 1990–2023: a systematic analysis for the Global

Burden of Disease Study 2023" *The Lancet Infectious Diseases* (Under Review).

7. GBD 2023 Type 1 Diabetes Collaborators (2025) "The global burden of type 1 diabetes: a systematic analysis for the Global Burden of Disease Study 2023." *Under Review*.
8. GBD 2023 Cervical Cancer Collaborators (2025) "Global, regional and national burden of cervical cancer and associated risk factors from 1990 to 2023 and forecasts to 2050: A systematic analysis for the Global Burden of Disease Study 202" *Under Review*.

C. Papers Under Review

1. Kundu, P., Shahariar, M., Munny, F. A., Riya, A. T., Lota, H. B., Ahmed, M. Z., Meme, M. A., Kona, U. M. P., Mahmuda, F., Chowdhury, S. S., Sraboni, S. A., & ***Hossain, M. J.** (2026) "Computational identification of *Taxus baccata*-derived phytochemicals targeting EZH2 to overcome therapeutic resistance in melanoma" *Scientific Reports*. <https://doi.org/10.21203/rs.3.rs-8494315/v1>
2. Islam, M. W., Rahman, M. M., Naznin, H., Hossain, M. S., Akter, T., Shatabde, Z. A., & ***Hossain, M. J.** (2025) "Integrative bioinformatics analysis reveals COL13A1 and COL23A1 as potential diagnostic and prognostic biomarkers in thyroid cancer" *Health Science Reports*.
3. Islam, M. W., Fariha, F. T. J., Ahmed, M. Z., Ferdous, J., Lota, H. B., Hossain, M. S., Kundu, P., Shahjahan, M., & ***Hossain, M. J.** (2025) "Bioinformatics-driven multi-omics profiling of CDK1 and CDK6 identifies prognostic and therapeutic roles in breast cancer" *Health Science Reports*.
4. Billah, M. M., Mabsurah, K., Ahammad, K., Yeana, I. J., Sumaiya, M., Islam, T., Bhattacharjee, A., Ferdous, J., & ***Hossain, M. J.** (2025) "Multi-omics pan-cancer analysis reveals an immunological role and prognostic potential of WDR76" *Discover Oncology*.
5. Sony, S. A., Kundu, L. R., Limon, M. H., Chowdhury, T. B. K., & ***Hossain, M. J.** (2025) "Predicting early antenatal care initiation at the first trimester among reproductive women in Bangladesh using machine learning" *BMC Pregnancy and Childbirth*.
6. ***Hossain, M. J.**, Shahariar, M., Barsha, L.H.J. Shahjahan, M., Towhid, ST., Sheikh, MK., Hasan, MM., Rahman, MH., Sazid, MS. (2025) "Lack of knowledge and training about antibiotic resistance among community pharmacists in Bangladesh: a cross-sectional study" *Health Science Reports*.
7. Billah, M. M., Mabsurah, K., Ahammad, K., Yeana, I. J., Sumaiya, M., Islam, T., Bhattacharjee, A., Ferdous, J., & ***Hossain, M. J.** (2025) "Multi-omics pan-cancer analysis reveals an immunological role and prognostic potential of WDR76"
8. ***Hossain, M. J.**, Shahariar, M., Barsha, L. H. J., Shahjahan, M., Towhid, S. T., Sheikh, M. K., Hasan, M. M., Rahman, M. H., & Sazid, M. S. (2024) "Lack of knowledge and training about antibiotic resistance among community pharmacists in Bangladesh: A cross-sectional study" *Health Science Reports*.

D. Working Papers

1. **Hossain, M. J.**, Dev, P.C. (2026) "Fezf2-mediated cortical development: Multi-omics analysis" *Working Paper*. <https://github.com/hossainlab/fezf2-multiomics>
2. **Hossain, M. J.**, Dev, P.C. (NA) "EWSAtlas: A Harmonized Single-Cell Transcriptomics Atlas of Human Ewing sarcoma" <https://github.com/hossainlab/EWSAtlas>
3. **Hossain, M. J.**, Dev, P.C., & Shahariar, M. (2026) "Deep Learning-Driven AMR Drug Repurposing" *Working Paper*. https://github.com/hossainlab/amr_drug_repurposing
4. **Hossain, M. J.** (2026) "DeepAMR: AI-powered antimicrobial resistance prediction from genomic data" *Working Paper*. <https://github.com/hossainlab/DeepAMR>

5. **Hossain, M. J.**, ****Tabassum, R.**, ****Ahmad, F.**, ****Ahamed, M. S.**, ****Aditee, L. M.**, **Fuad, M.**, **Shahjahan, M.** (2025) "The Pan-Cancer Consensus Transcriptome (PCCT): A Multi-Layered Meta-Analysis and Machine Learning Framework for Cross-Platform Biomarker Discovery" *Working Paper*. https://github.com/gsabioinfointernship/Pancreatic_Cancer_Meta_Analysis/

Presentations

A. Invited Speaker

1. "Undergraduate Research - Importance, Benefits, and Challenges". CHIRAL Bangladesh. Dhaka, Bangladesh. Mar 21, 2019.
2. "State the Art of Microbial Genome Analysis". Jagannath University. Dhaka, Bangladesh. May 21, 2018.
3. "Computational Biology and Bioinformatics Research in Resource-Limited Settings: Strategies, Tools and Opportunities". Jagannath University Higher Study and Research Society. Dhaka, Bangladesh. Jun 25, 2015.
4. "Mastering Biomedical Data Management". IFMSA Bangladesh. Dhaka, Bangladesh. Apr 21, 2014.

B. Conference Presentations

1. "Fezf2-Mediated Cortical Development: Multi-Omics Single-Cell Analysis", with ***Islam, MM.** Network of Young Biotechnologists Bangladesh. Dhaka, Bangladesh. Apr 20, 2026.
2. "Diagnostic and Electrophysiological Features of Hirayama Disease in Young Adult Male: A Case Report", with Manisha Das. Institute of Epidemiology, Disease Control and Research (IEDCR). Dhaka, Bangladesh. May 21, 2020.
3. "Quantitative Microbial Risk Assessment from Vancomycin-Resistant *Enterococcus faecalis* and *Enterococcus faecium* from a Specific Neighborhood in Dhaka City, Bangladesh", with Nayeem, M.U., Mrityika, M.A., Azad, A.K., Ferdous, J., Ahmed, S., Sanyal, S.K., Towhid, S.T.. Bangladesh Society of Microbiologists. Dhaka, Bangladesh. May 21, 2019.

C. Posters

1. "Identification of Subtype-Specific Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis", with ****Ahmad, F.**, **Fuad, M.**, **Shahjahan, M.**, & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
2. "Network-Driven Bioinformatics Approach Uncovers Prognostic Signatures and Therapeutic Targets in Pancreatic Ductal Adenocarcinoma", with ****Aditee, L. M.**, **Fuad, M.**, **Shahjahan, M.**, & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
3. "Bioinformatic Analysis of Transcriptomic Data Reveals Molecular Signatures Driving Colorectal Cancer", with ****Ahamed, M. S.**, **Fuad, M.**, **Shahjahan, M.**, & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.
4. "Comparative Transcriptomic Meta-Analysis Reveals Shared and Distinct Transcriptional Networks and Pathway Crosstalk in Hepatocellular Carcinoma for Biomarker and Therapeutic Target Identification", with ****Tabassum, R.**, **Fuad, M.**, **Shahjahan, M.**, & **Hossain, M. J.**. Bangladesh Society of Microbiologists. University of Dhaka, Bangladesh. Dec 25, 2025.

Media Coverage

1. May 27, 2024, **The Business Standard**: Quoted in article [What the rise in self-medication tells us about the country's healthcare system](#).
2. Feb 08, 2024, **The Business Standard**: Study featured in article ["Parental lack of antibiotic knowledge imperils child health in Bangladesh: Study"](#).